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TECHNICAL REPORT

System Reference document (SRdoc); Short Range Devices (SRD) using Ultra Wide Band (UWB); Technical characteristics and spectrum requirements for indoor security scanners within 12 GHz - 40 GHz

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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee Electromagnetic compatibility and Radio spectrum Matters (ERM).

Modal verbs terminology

In the present document "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Introduction

The present document includes necessary information to support the co-operation under the MoU between ETSI and the Electronic Communications Committee (ECC) of the European Conference of Post and Telecommunications Administrations (CEPT).

The present document covers the request for harmonized spectrum for sensor or radiodetermination applications using millimetre-wave technology within the frequency range 12 GHz - 40 GHz. Communications applications or hybrid applications as a combination of sensor and communications applications are not treated within the scope of the present document.

The intention of the production of the present document is to create a basis for the industry to facilitate the market launch of new innovative and useful security scanners while avoiding any harmful interference with other radio services and equipment.

1 Scope

The present document describes security scanner applications installed and operated indoors within the frequency range 12 GHz - 40 GHz which may require a change of the present frequency designation/utilization within CEPT.

The present document includes in particular:

- market information;
- technical information;
- regulatory issues.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding but are not required for conformance to the present document.

[i.1] CEPT ECC Report 139 (02-2010): "Impact of Level Probing Radars Using Ultra-Wideband Technology on Radiocommunications Services".

[i.2] ETSI EN 302 372 (V2.1.1) (2016-12): "Short Range Devices (SRD); Tank level Probing Radar (TLPR) equipment operating in the frequency ranges 4,5 GHz to 7 GHz, 8,5 GHz to 10,6 GHz, 24,05 GHz to 27 GHz, 57 GHz to 64 GHz, 75 GHz to 85 GHz; Harmonised Standard covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.3] ETSI EN 302 729 (V2.1.1) (2016-12): "Short Range Devices (SRD); Level Probing Radar (LPR) equipment operating in the frequency ranges 6 GHz to 8,5 GHz, 24,05 GHz to 26,5 GHz, 57 GHz to 64 GHz, 75 GHz to 85 GHz; Harmonised Standard covering the essential requirements of article 3.2 of the Directive 2014/53/EU".

[i.4] ETSI TS 103 361 (V1.1.1) (2016-03):" Short Range Devices (SRD) using Ultra Wide Band technology (UWB); Receiver technical requirements, parameters and measurement procedures to fulfil the requirements of the Directive 2014/53/EU".

[i.5] ETSI EN 305 550-1 (V1.2.1) (2014-10): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Short Range Devices (SRD); Radio equipment to be used in the 40 GHz to 246 GHz frequency range; Part 1: Technical characteristics and test methods".

[i.6] ETSI EN 303 883 (V1.1.1) (2016-09): "Short Range Devices (SRD) using Ultra Wide Band (UWB); Measurement Techniques".

[i.7] ETSI EN 305 550-2 (V1.2.1) (2014-10): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Short Range Devices (SRD); Radio equipment to be used in the 40 GHz to 246 GHz frequency range; Part 2: Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive".

[i.8] CEPT/ERC/Recommendation 74-01: "Unwanted emissions in the spurious domain", May 2019 edition.

[i.9] ERC/REC 70-03: "Relating to the use of Short Range Devices (SRD)", June 2019.

[i.10] EFIS database.

[i.11] ECC Report 023: "Compatibility of automotive collision warning Short Range Radar operating at 24 GHz with FS, EESS and Radio Astronomy", May 2003.

[i.12] ECC Report 166: "Coexistence between zenith-pointing meteorological radars at 24 GHz and 35 GHz and systems in other radio services", May 2011.

[i.13] ETSI EN 303 940-1 (V1.1.1): "Short Range Devices (SRDs) using Ultra Wide Band technology (UWB); Harmonised standard for access to radio spectrum; Part 1: Indoor Millimeter Wave Security Scanners operating in the band 69,8 GHz to 80,5 GHz".

[i.14] ETSI EN 303 940-2: "Short Range Devices (SRDs) using Ultra Wide Band technology (UWB); Harmonised Standard for access to radio spectrum: Security Scanning devices; Part 2: MicroWave Security Scanners operating < 12 GHz".

[i.15] Report Recommendation ITU-R SM.2092-0 (2007): "Studies related to the impact of active services allocated in adjacent or nearby bands on Earth exploration-satellite service (passive)".

[i.16] Report Recommendation ITU-R SM.2091-0 (2007): "Studies related to the impact of active space services allocated in adjacent or nearby bands on radio astronomy service".

[i.17] ECC(22)039: "Draft CEPT Briefing Note on WRC-23 Agenda Item 1.16".

[i.18] ETSI TR 102 601 (V1.1.1) (2007-12): "Electromagnetic compatibility and Radio spectrum Matters (ERM); System reference document; Short Range Devices (SRD); Equipment for Detecting Movement using Ultra Wide Band (UWB) radar sensing technology; Level Probing Radar (LPR)-sensor equipment operating in the frequency bands 6 GHz to 8,5 GHz; 24,05 GHz to 26,5 GHz; 57 GHz to 64 GHz and 75 GHz to 85 GHz".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI EN 303 883 [i.6], ETSI TS 103 361 [i.4] and the following apply:

Activity Factor (AF): ratio of active measurement periods t_m (bursts, sweeps, scans) within the overall repetitive measurement cycle T_m

algorithm: automatic detection software

avatar: anonymous representation of a human figure for locating the detected objects

body position: bearing, including bearing of arms and legs

Frequency Modulated Continuous Wave (FMCW): periodically linear frequency sweep of the transmit signal

NOTE: For distance measurement sensors often a sawtooth or a triangular modulation scheme is used. By mixing the current transmit signal with the reflected signal the round-trip time of the individual echoes and thus the distance of the different targets can be determined.

front panel: panel in front of the screened individual

inspector: person who performs screening using the scanner or who carries out manual screening

manual screener: inspector who performs manual screening

manual screening: inspection by passing the hands over the body panel: plate for scanning the front and back of the screened individual

NOTE: See also "front panel" and "rear panel".

position: place where the scanned person stands for screening

posture: body position

rear panel: panel behind the screened individual

resolution station: area in which manual screening is performed

resolution station monitor: touchscreen used by the manual screener at the resolution station

scan procedure: sequence of actions to be performed by the scanned person and the screener

scanner: equipment used for screening

screened individual: person to be screened

screener: inspector who operates the scanner(s)

screening: checking of the screened individual

security checkpoint: location at which the security check is performed

security scanner: equipment used for screening

Stepped Frequency Continuous Wave (SFCW): transmit frequency in an SFCW modulation scheme not swept in a linear manner but rather in a stepped way with defined frequency increments and with a certain dwell time on each individual frequency step

touchscreen: screen for controlling the scanner and for displaying the scan results

3.2 Symbols

For the purposes of the present document, the following symbols apply:

f_L	lowest frequency of the operating bandwidth
f_H	highest frequency of the operating bandwidth
t_m	active measurement time segment
T_m	overall repetitive measurement cycle time
Rx	Receiver

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AZ/EL	AZimut/ELevation
CW	Continuous Wave
DAA	Detect And Avoid
DUT	Device Under Test
EESS	Earth Exploration Satellite Service
EIRP	Equivalent Isotropically Radiated Power
EM	ElectroMagnetic
ESIM	Earth Station In Motion
FMCW	Frequency Modulated Continuous Wave
FS	Fixed Service
FSK	Frequency Shift Keying

FSS	Fixed Satellite Service
FWA	Fixed Wireless Access
GPR/WPR	Ground Probing and Wall Probing Radar
GSO	Geostationary Orbit
HERP	Hazards of Electromagnetic Radiation to Personnel
ITU-R	International Telecommunication Union - Radiocommunication Sector
LDC	Low Duty Cycle
LPR	Level Probing Radar
MIMO	Multiple Input Multiple Output
MSS	Microwave Sub System
MWSSc	Microwave Security Scanner
NGSO	Non Geo-Stationary Orbit
ODC	Object Discrimination and Characterization
OJEU	Official Journal of the European Union
PAX	Person to be checked by the SSc/MWSSc
PMP	Point to Multi-Point
PMSE	Programme Making and Special Events
Q-MSS	Q Microwave SubSystem
QPS	Quick Personal Security scanner
RAS	Radio Astronomy Service
RF	Radio Frequency
RNSS	Radio Navigation-Satellite Service
SFCW	Stepped Frequency Continuous Wave
SRD	Short Range Devices
SRR	Short Range Radar
SSc	Security Scanner
TC	Technical Committee
TGUWB	Task Group Ultra WideBand
UWB	Ultra WideBand

4 Comments on the System Reference Document

4.1 Statements by ETSI Members

No statements or comments have been issued by ETSI members.

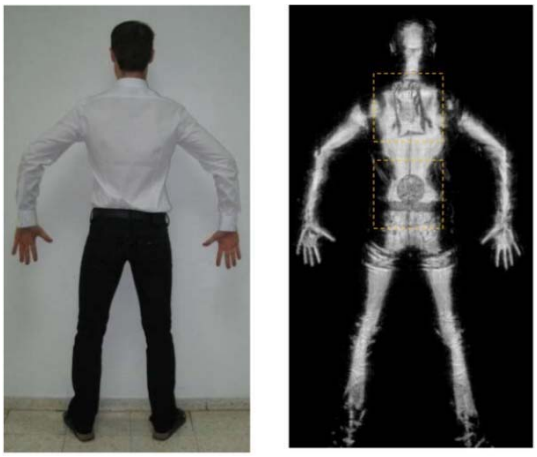
5 Presentation of the system and technology

5.1 Introduction

A Security Scanner (SSc) is a device that detects objects on a person's body for security screening purposes, without making any physical contact.

SScs work with electromagnetic waves. They use wavelengths in the millimetre range, e.g. around (12 - 40 GHz) and operate at low signal power. The output power of an SSc, when transmitting, is in the very low milliwatt domain (e.g. 1 mW to 10 mW) and rather low compared to other radars or communication devices. This is much lower than, for example, that of Wireless Local Area Networks (WLAN) or that of the ubiquitous mobile phone. The output power for WLAN is typically around 100 mW to 200 mW. A regular mobile phone transmits at about 1 000 mW to 2 000 mW. These output powers are therefore around 100 to 1 000 times higher than an SSc.

Millimetre waves penetrate human skin only to a small extent and only by a few fractions of a millimetre as shown in Figure 1. They are for the most part reflected by the surface of the skin. Basically, the depth by which electromagnetic waves penetrate human skin decreases with increasing frequency. In the said frequency range around 30 GHz, the penetration depth is much less than for wavelengths of 150 mm to 300 mm, i.e. at frequencies of around 1 GHz - 2 GHz which are used by mobile phones. For fundamental physical reasons, millimetre waves are unable to penetrate the inside of the human body.



NOTE: Picture courtesy of Rohde & Schwarz and permission to be used in the present document has been granted.

Figure 1: Exemplary measurement data of an SSc

Generally, the length of time which the above-mentioned power is emitted is short (e.g. only a couple of milliseconds). An SSc is not transmitting permanently, but it is active for a short time only, immediately after the screener triggers a scan. Compared to the waves used by mobile phones, the signals emitted by the scanner penetrate less deeply (about one tenth), have a much lower power (about one thousandth) and are active for a considerably shorter time (about one ten thousandth). With the exception of the period of time mentioned above, an SSc is inactive and does not emit a signal.

Some SSc can digitally exclude specific frequency ranges to avoid restricted bands. Within the usable bandwidth, the SSc is capable of dynamically selecting its operating frequencies, providing flexibility for different imaging scenarios. Individual frequency points can be omitted as needed, ensuring compliance with regulatory constraints while maintaining high-resolution millimetre-wave imaging performance.

5.2 Object detection and localization

SScs are imaging devices used for detecting objects concealed on a person, on individual body parts, within clothing, or within items carried close to the body. Potentially dangerous or prohibited objects are detected by means of a measurement-based evaluation of millimetre-wave signals emitted by the scanner and backscattered by the person, their clothing, or by objects worn openly or concealed on or beneath clothing. These signals are received and processed by the scanner.

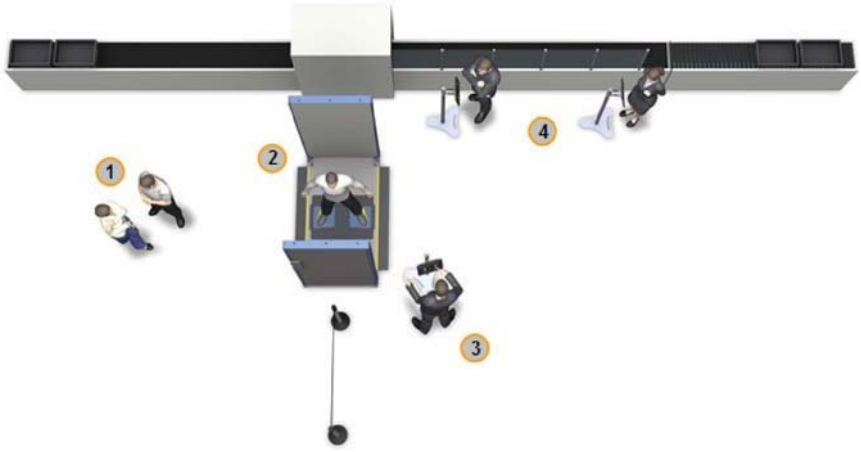
If necessary, the detection result is provided in the form of a visual alarm that allows precise localization of the detected objects by the operator. Typical applications for this technology include the detection of items relevant to security screening, commercial loss prevention, smuggling detection, and the inspection of persons, carried items, parcels, goods, or products in controlled environments such as government buildings or airport security checkpoints.



NOTE: Picture courtesy of Leidos and permission to be used in the present document has been granted.

Figure 2: Example for an SSc in operation at an airport

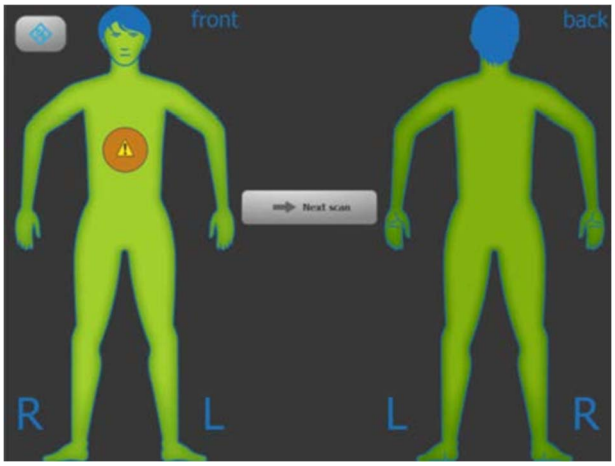
An SSc is usually used for screening persons at security checkpoints. The SSc automatically detects metallic and non-metallic objects of interest of all types concealed inside or beneath clothing. These objects include potentially dangerous items such as weapons, explosives and liquids or other objects which the device operating software is designed to detect.



NOTE 1: Picture courtesy of Rohde & Schwarz and permission to be used in the present document has been granted.
NOTE 2: 1) Two persons waiting to be scanned
2) A person being scanned
3) An inspector operating the SSc via a touchscreen
4) Further inspectors at resolution stations

Figure 3: Typical layout of a security checkpoint utilizing an SSc

Figure 3 shows an example layout of a security checkpoint equipped with an SSc, at an airport. On the left of Figure 3, a number of as yet unscreened screened individuals (1) are waiting in front of the SSc. The SSc here basically consists of two panels, the front panel and the back panel. A screened individual (2) has already entered the SSc and adopted the posture required for scanning, i.e. facing the front panel with arms slightly splayed. The inspector (3) at the scanner (screener) instructs the screened individual and triggers the scan at his touchscreen. He receives the scan results on the screen, e.g. simply a PASS, in case that no object had been detected, or an object indication as shown in Figure 4.

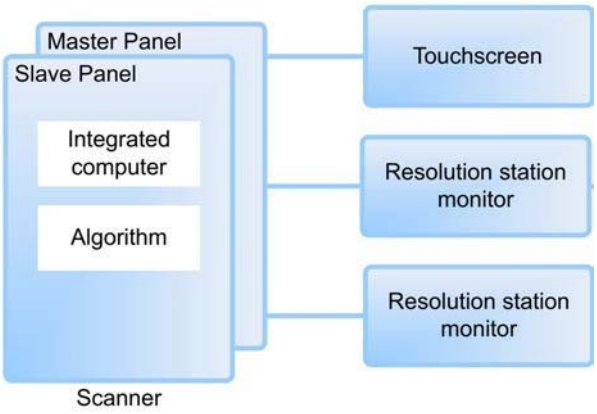


NOTE: Picture courtesy of Rohde & Schwarz and permission to be used in the present document has been granted.

Figure 4: Exemplary display on the touchscreen indicating a detected object

Optionally, the security checkpoint can have resolution stations. In Figure 3, one female and one male inspector (4) are waiting for screened individuals who may require manual screening. Each inspector has their own resolution station monitors, which are used to forward information (detection result) from the scanner to the resolution station and also to send information from the resolution stations back to the screener (availability notification).

Figure 5 shows the basic architecture of a security checkpoint as described above, consisting of the Security Scanner (SSc) with its two panels as the most important part, as well as peripheral components such as secondary monitors in addition to the touchscreen used for operation of the SSc.



NOTE: Picture courtesy of Rohde & Schwarz and permission to be used in the present document has been granted.

Figure 5: Typical architecture of a security checkpoint utilizing an SSc

6 Market Information

6.1 Scenarios and deployment

SScs enhance security in areas where threats from people may arise. SScs are not deployed in an outdoor setting and are used by trained personnel.

There are scenarios and safety zones which require SScs, such as:

1) Inside airports buildings or central transportation centres for example underground rail stations

2) Loss prevention for factories, data centres, authorities, etc.

3) Military facilities

4) Prisons

5) Government offices and courts

6) Concert halls or museum

7) VIP offices, and visiting sites

8) Inside indoor security check areas of stadiums & major sport sites and events

All considered buildings could be seen as a thermally efficient building.

SScs enable automated security screening of large numbers of people in a limited period of time. As such they are extremely useful for use at (public) locations that are frequented by large numbers of people who need to be screened in a limited time period. By installing and using SScs, the operator of the location can quickly detect concealed objects of interest and have an operator investigate these objects.

Air Transport Security Screening where passenger security checkpoints within terminal buildings of airports in controlled indoor areas with restricted public access and environmental protection (temperature-controlled halls).

Border Control and Customs Facilities where indoor customs halls and immigration control areas at international transport hubs in enclosed terminal buildings or dedicated inspection facilities with controlled access.

Stadiums and concert halls have been the target of many terror attacks in the past. To ensure that people cannot enter the site with any harmful or dangerous goods or tools like guns, knives or even explosives, SScs are highly needed to scan the people before entering the event. Event venues such as indoor arenas, conference centres, and exhibition halls also need to be secured with permanent or semi-permanent screening checkpoints located within enclosed structures.

Government offices as well as major court houses could install and use SScs to protect their staff as well as other people who are in the building. People entering such sites might be infuriated about their reason to go to government or court houses. Therefore, they could be a high danger for the staff and others. Critical infrastructure sites requiring indoor access control points include government buildings, courthouses, nuclear facilities, and research centres and will typically have secured indoor lobbies or screening vestibules with physical access barriers using SScs.

The past has shown that assassins choose locations with a crowd of people or locations with specific persons, like celebrities or persons with a specific role like a lawyer. In order to avoid it, people visiting such sites should be checked through SScs which will increase safety.

SScs ensure loss prevention in sensitive environments such as factories, data centres, and government facilities. It prevents unauthorized items from being smuggled in or out, including data storage devices, confidential information, unreleased products, and valuable goods.

Using SScs maintains the security of the location while delivering an acceptable screening time. Screening time is relevant for access to the location, for example a security lane at an airport is handling significant amounts of people in a short period of time. Reducing the screening time will work towards making the airport accessible for departing airline screened individuals.

All applications and building typologies share the requirement for a high level of security or include areas that demand increased security standards. The buildings are characterized by enclosed architectural form, are predominantly constructed of concrete with a closed roof and deliberately avoid open or permeable spatial layouts due to security issues. This closed and self-contained thermal efficiency building design and the controlled internal conditions, clearly distinguishing these typologies from traditional building forms or outside spaces. All checkpoints follow a similar process as shown in the security checkpoint example. At airports there may be several security lines with numbers ranging between typically 3 (smaller airports) to 10 (bigger airports with multiple terminal buildings). Scanners are deployed in a proximity of 10 m - 20 m to each other as the devices are rather bulky and the security screening process requires space. For critical infrastructure security screening in other venues often a single scanner is sufficient.

Multiple SScs do not operate in a synchronized manner. Signals emitted from SScs are typically at the time of trigger when there is someone inside the scanner, which means scanners will operate independently with only a subset transmitting at the same time. Therefore, even with multiple scanners in operation at the same location transmit power will most likely not entirely aggregate, and even when scanners are in operation at the same time it is highly unlikely multiple scanners will be transmitting on the same sub-frequency (i.e. at the same part of the frequency sweep) simultaneously.

6.2 Requirements and increasing threat

The present document describes typical system specifications and use cases at the time it was written. Future development is expected yield critical market benefits which may include but are not limited to improved performance for enhanced security, reduced burden to security officers, and increased throughput rates with modern concept of operations as applicable to security checkpoints.

SScs work in a number of frequency bands including 3 - 10 GHz and 70 GHz. This additional band is needed because it is less affected by clothing than the 60 - 90 GHz band and so can deliver more accurate security scanning which has benefits both in detecting threats but also in reduced false positives with lower numbers of manual searches, saving time and inconvenience. Using higher frequencies than 10 GHz increases the resolution, allowing smaller items to be accurately detected and again reducing false positives.

The layout and physical dimension of SScs in this frequency band combines the advantages of high frequency scanners in 69,8 - 80,5 GHz and lower frequency scanners in 3,6 - 10,6 GHz. At high frequency the antenna clusters can be physically small but still offer high angular resolution due to the high number of antennas. At lower frequency there are less antennas and resolution is less. Penetration and effects by clothing is also less at low frequency and greater at high frequencies.

An increasing demand to detect and screen shoes is of major importance and requires additional solutions where current systems performance lacks. The combination of a high resolution in range, azimuth and elevation with penetration through thick material like shoes is required. This is done by signal bandwidth and the size of the antenna array and frequencies in the range of 12 - 40 GHz.

7 Technical information

7.1 Example specifications of an SSc

There are different SScs with different form-factors for different security applications. Some are used for screening the entire body of the person at one scan. Other SScs are used to scan only parts of the body or scan for material inside packages. All SScs mentioned in the present document operate in a similar frequency range of 12 - 40 GHz, have low output peak power in common, while their number of antennas and physical dimension and format differs depending on the scanning application and manufacturer layout.

7.2 Summary specifications of a Leidos scanner (SSc 1)

As an example, the RF specifications of a Leidos scanner (SSc 1) are given.

Table 1: RF specifications of SSc 1

Attribute	Value
Number of transmitting antennas per mast	108
Number of receiving antennas per mast	72
Antenna raster	4,6 mm (vertical)
Aperture (W × H) of a swept mast	1 280 mm × 2 000 mm
Beam width of an antenna (horizontal and vertical)	60°
Antenna tilt	0°
Transmitting time of a single antenna at a single frequency	51 ns
Transmitting time of a mast at a single frequency	11 µs
Total transmitting time per posture per mast	674 ms
Lag between front and back scan	< 100 ms
Repetition rate	0,16 Hz
Start frequency	20 GHz
Stop frequency	40 GHz
Peak output power	< 0 dBm
Antenna gain	12 dBi
Polarization	Circular

The scanner repeatedly sweeps a chirped unmodulated sine-wave Frequency Modulated Continuous Wave (FMCW) signal over the frequency range 20 - 40 GHz. Figure 6 below depicts timing of sweeps and transmission of the wave. As shown:

a) The minimum possible time between scans is approximately 4 seconds but, depending upon checkpoint loading and the time required by the operator to clear alarms, the time between scans is typically 10 seconds or more.

b) As the antenna mast rotates through the nominal 110-degree mechanical travel, vertical scan lines are triggered at equidistant intervals around the circumference of the mast path. The system remains idle with the transmitter disabled until the system operator initiates the next scan sequence.

c) A full subject scan data capture sequence takes 1,3 seconds, followed by a pause to process the image and analyse the results.

d) The chirped FMCW signal is repeatedly swept over the frequency range 20 - 40 GHz at a rate of 1,46 MHz/ns. This chirp is repeated at each of the transmit antenna sampling positions along the height of the mast during a vertical scan line data capture sequence.

e) The transmitted waveform, or chirp, over the 20 - 40 GHz band takes 13,7 µs.

f) There is a 0,5 µs period with the transmitter disabled between chirps to switch to the next sampling point on the mast resulting in a total chirp repetition period of 14,2 µs, as shown in Figure 6.

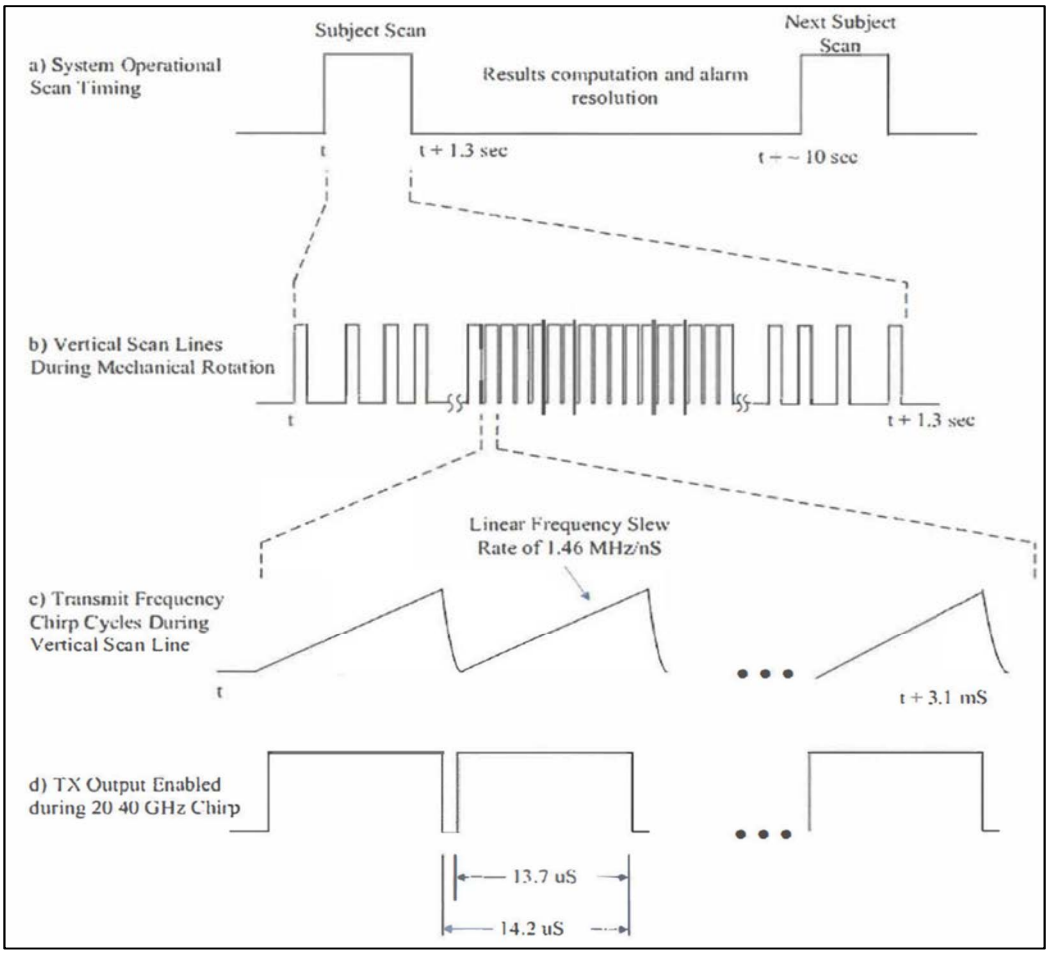


Figure 6: Scanning and Transmission Timings

7.3 Summary specifications of an R&S SSc (SSc 2)

As a second example, the RF specifications of an SSc 2 from Rohde & Schwarz are given below:

Table 2: RF specifications SSc 2

Attribute	Value
Number of transmitting antennas per panel	1 152
Number of receiving antennas per panel	1 152
Number of antenna tiles per SSc	8
Number of transmitting antennas per tile	144
Number of receiving antennas per tile	144
Antenna raster (TX)	8 / 24 mm
Antenna raster (RX)	16 mm
Aperture (W x H) of a panel	36,8 mm x 36,8 mm (2 x 2 tiles, 2 per system)
Antenna height above ground level	15 mm
Beam width of an antenna (horizontal and vertical)	60°
Number of frequencies	192
Start frequency	12 GHz
Stop frequency	40 GHz
Peak output power	5 dBm
Antenna gain	8 dBi
Polarization	linear

SSc 2 transmits a Frequency Shift Keying (FSK) signal between a lower f_L and upper f_H frequency. In between those frequencies several frequencies (e.g. 192) are transmitted in a Continuous Wave (CW) manner. The t_{on}, t_{hop} and t_{off} times depend on the scanning nature. The SSc 2 FSK signal allows to avoid transmission at dedicated frequencies or step beyond as mitigation technique.

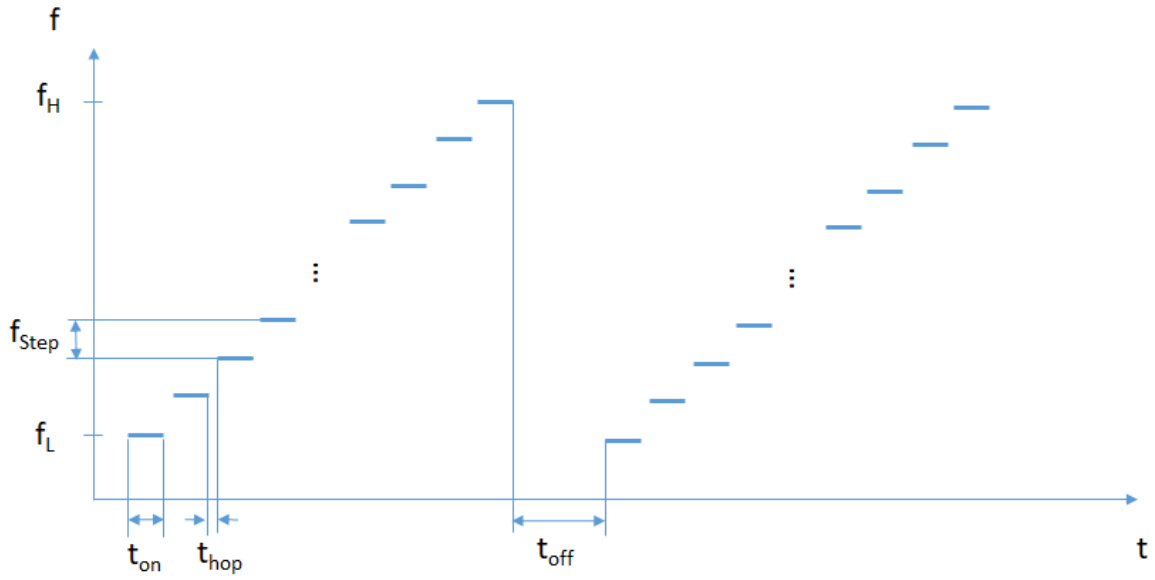


Figure 7: FSK Signal of an exemplary SSc 2

7.4 Wanted and unwanted emissions

7.4.1 Transmitter (wanted) emissions within the operating frequency band

The datasheet values are given in the RF specification in Table 1 and Table 2.

7.4.2 Transmitter (unwanted) emissions outside the operating frequency band

The transmitter unwanted emissions are emissions from the SSc originating from the transmitter but outside the operating band including above the band and below the band. It is the average power per unit bandwidth (centred on that frequency) radiated in the direction of main radiation (main lobe of the antenna). They are measured as maximum mean power spectral density (specified as EIRP) in the direction of main radiation of the radio device under test usually over a frequency range from 30 MHz to 2 times the carrier frequency except for the operating band of the SSc.

The limit for the transmitter unwanted emissions outside the operating bandwidths is proposed to be 20 dB less than the maximum mean power spectral density within the operating bandwidths. This value is adopted from ETSI EN 302 729 [i.3], clause 4.3.8 which states that *"For LPR operating in one of these operating bandwidths the maximum value of mean power spectral density (dBm/MHz) shall be 20 dB less than the in-band density specified in table 3. For LPR operating within the 24,05 GHz to 26,5 GHz band, the unwanted emissions in the 23,6 GHz to 24,0 GHz "passive band" shall be at least 30 dB less than the in-band limits specified in table 3."*

7.4.3 Other (unwanted) transmitter emissions

Transmitters often emit very low power radio signals, comparable with the power of spurious emissions from digital and analogue circuitry. If it can be clearly demonstrated that an emission from the SSc is not a transmitter emission (e.g. by disabling the SSc's transmitter) or it can clearly be demonstrated that it is impossible to differentiate between other emissions and the transmitter (unwanted) emissions, that emission or aggregated emissions should be considered against the other emission limits.

The proposed limits for other emissions can be found in Table 3. The limits are in line with the recommended values in CEPT/ERC/REC 74-01 [i.8].

Table 3: Proposed limits for other emissions

Frequency range	Limit
30 MHz to 1 GHz	-57 dBm (EIRP)
above 1 GHz	-47 dBm (EIRP)

7.4.4 Receiver spurious emissions

Receiver spurious emissions are emissions of the receiver of a DUT when it is in a receive-only mode or is a receive-only device. Consequently, receiver spurious emission testing should apply only when the equipment can work in a receive-only mode or is a receive-only device.

For collocated Tx/Rx equipment that does not have a receive-only mode, the receiver spurious emissions are usually considered within the scope of the transmitter parameter "other emissions".

The proposed limits for receiver spurious emissions can be found in Table 4. The limits are in line with the recommended values in CEPT/ERC/REC 74-01 [i.8].

Table 4: Proposed limits for receiver spurious emissions

Frequency range	Limit
30 MHz to 1 GHz	-57 dBm (EIRP)
above 1 GHz	-47 dBm (EIRP)

7.5 Current ITU and European Common Allocations

The following ITU Region 1 and EC allocations apply [i.10].

Table 5: Current frequency allocations in the requested bands

11,7 GHz - 12,5 GHz	Allocations: BROADCASTING-SATELLITE, MOBILE EXCEPT AERONAUTICAL MOBILE Applications: Broadcasting (satellite), ESIM, HEST, NGSO FSS, AES
12,5 GHz - 12,75 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), FIXED-SATELLITE (SPACE-TO-EARTH) Applications: AES, ESIM, ESV, NGSO FSS, HEST
12,75 GHz - 13,25 GHz	Allocations: FIXED, FIXED-SATELLITE (EARTH-TO-SPACE) Applications: Fixed, FSS Earth stations, AES
13,25 GHz - 13,4 GHz	Allocations: AERONAUTICAL RADIONAVIGATION, EARTH EXPLORATION-SATELLITE (ACTIVE), SPACE RESEARCH (ACTIVE) Applications: Active sensors (satellite), Airborne doppler navigation aids, Maritime radar
13,4 GHz - 13,65 GHz	Allocations: EARTH EXPLORATION-SATELLITE (ACTIVE), RADIOLOCATION, SPACE RESEARCH, FIXED-SATELLITE (SPACE-TO-EARTH) Applications: Active sensors (satellite), Airborne doppler navigation aids, FSS Earth stations, Radiodetermination applications, Radiolocation (military), Maritime radar
13,65 GHz - 13,75 GHz	Allocations: EARTH EXPLORATION-SATELLITE (ACTIVE), RADIOLOCATION, SPACE RESEARCH Applications: Active sensors (satellite), Airborne doppler navigation aids, Radiodetermination applications, Radiolocation (military), Maritime radar
13,75 GHz - 14 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), RADIOLOCATION, Space research Applications: FSS Earth stations, Maritime radar, Passive sensors (satellite), Radiodetermination applications, Radiolocation (military), -
14 GHz - 14,25 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), Space research, Mobile-Satellite (Earth-to-space) Applications: AES, ESV, HEST, MSS Earth stations, NGSO FSS, VSAT, ESIM
14,25 GHz - 14,3 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), Mobile-Satellite (Earth-to-space), Space research Applications: AES, ESIM, ESV, MSS Earth stations, NGSO FSS, VSAT
14,3 GHz - 14,4 GHz	Allocations: Mobile-Satellite (Earth-to-space), FIXED-SATELLITE (EARTH-TO-SPACE) Applications: AES, ESV, FSS Earth stations, MSS Earth stations, NGSO FSS, VSAT, ESIM
14,4 GHz - 14,47 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), Mobile-Satellite (Earth-to-space) Applications: AES, ESIM, ESV, FSS Earth stations, MSS Earth stations, NGSO FSS, VSAT
14,47 GHz - 14,5 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), Mobile-Satellite (Earth-to-space), Radio astronomy Applications: AES, ESV, FSS Earth stations, MSS Earth stations, NGSO FSS, Radio astronomy, VSAT, ESIM
14,5 GHz - 14,75 GHz	Allocations: FIXED, MOBILE, Radio astronomy Applications: Aeronautical military systems, Fixed, Land military systems, Maritime military systems, Radio astronomy
14,75 GHz - 14,8 GHz	Allocations: FIXED, MOBILE, Radio astronomy Applications: Aeronautical military systems, Land military systems, Maritime military systems, Radio astronomy
14,8 GHz - 15,35 GHz	Allocations: FIXED, MOBILE, Radio astronomy Applications: Aeronautical military systems, Fixed, Land military systems, Maritime military systems, Radio astronomy
15,35 GHz - 15,4 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE) Applications: Passive sensors (satellite), Radio astronomy
15,4 GHz - 15,41 GHz	Allocations: Aeronautical radionavigation, RADIOLOCATION Applications: Airborne doppler navigation aids, Radiolocation (civil)
15,41 GHz - 15,43 GHz	Allocations: RADIOLOCATION, Aeronautical radionavigation Applications: Airborne doppler navigation aids, Radiolocation (civil)
15,43 GHz - 15,63 GHz	Allocations: AERONAUTICAL RADIONAVIGATION, RADIOLOCATION, FIXED-SATELLITE (EARTH-TO-SPACE) Applications: Airborne doppler navigation aids, FSS Earth stations, Radiolocation (civil)

15,63 GHz - 15,7 GHz	Allocations: RADIOLOCATION, AERONAUTICAL RADIONAVIGATION Applications: Airborne doppler navigation aids, Radiolocation (civil)
15,7 GHz - 16,6 GHz	Allocations: RADIOLOCATION Applications: Radiolocation (military)
16,6 GHz - 17,1 GHz	Allocations: RADIOLOCATION, Space research (deep space) (Earth-to-space) Applications: Radiolocation (military)
17,1 GHz - 17,2 GHz	Allocations: RADIOLOCATION, Mobile Applications: Radiolocation (military), GBSAR
17,2 GHz - 17,3 GHz	Allocations: EARTH EXPLORATION-SATELLITE (ACTIVE), MOBILE, RADIOLOCATION, SPACE RESEARCH (ACTIVE) Applications: GBSAR, Radiolocation (military)
17,3 GHz - 17,7 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), Radiolocation, FIXED-SATELLITE (SPACE-TO-EARTH) Applications: Feeder links, GSO ESOMPs, NGSO ESOMPs, Radiolocation (military), FSS Earth stations
17,7 GHz - 18,1 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), FIXED-SATELLITE (SPACE-TO-EARTH), FIXED Applications: Feeder links, Fixed, FSS Earth stations, NGSO ESOMPs, GSO ESOMPs
18,1 GHz - 18,4 GHz	Allocations: FIXED, FIXED-SATELLITE (EARTH-TO-SPACE), FIXED-SATELLITE (SPACE-TO-EARTH), METEOROLOGICAL-SATELLITE (SPACE-TO-EARTH), INTER-SATELLITE Applications: Feeder links, Fixed, GSO ESOMPs, NGSO ESOMPs, FSS Earth stations
18,4 GHz - 18,6 GHz	Allocations: FIXED, INTER-SATELLITE, FIXED-SATELLITE (SPACE-TO-EARTH) Applications: Fixed, FSS Earth stations, GSO ESOMPs, NGSO ESOMPs
18,6 GHz - 18,8 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), FIXED, FIXED-SATELLITE (SPACE-TO-EARTH) Applications: Fixed, FSS Earth stations, GSO ESOMPs, NGSO ESOMPs, Passive sensors (satellite)
18,8 GHz - 19,3 GHz	Allocations: FIXED, FIXED-SATELLITE (SPACE-TO-EARTH), INTER-SATELLITE Applications: Fixed, FSS Earth stations, GSO ESOMPs, NGSO ESOMPs
19,3 GHz - 19,7 GHz	Allocations: FIXED, FIXED-SATELLITE (SPACE-TO-EARTH) (EARTH-TO-SPACE) Applications: Fixed, GSO ESOMPs, NGSO ESOMPs, FSS Earth stations
19,7 GHz - 20,1 GHz	Allocations: FIXED-SATELLITE (SPACE-TO-EARTH), Mobile-Satellite (space-to-Earth), INTER-SATELLITE Applications: FSS Earth stations, GSO ESOMPs, HEST, MSS Earth stations, NGSO ESOMPs
20,1 GHz - 20,2 GHz	Allocations: FIXED-SATELLITE (SPACE-TO-EARTH), INTER-SATELLITE, MOBILE-SATELLITE (SPACE-TO-EARTH) Applications: FSS Earth stations, GSO ESOMPs, MSS Earth stations, NGSO ESOMPs, HEST
20,2 GHz - 21,2 GHz	Allocations: FIXED-SATELLITE (SPACE-TO-EARTH), MOBILE-SATELLITE (SPACE-TO-EARTH) Applications: MSS Earth stations, Satellite systems (military)
21,2 GHz - 21,4 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), FIXED, MOBILE, SPACE RESEARCH (PASSIVE) Applications: Video PMSE
21,4 GHz - 22 GHz	Allocations: BROADCASTING-SATELLITE Applications: Broadcasting (satellite), Video PMSE, SRR
22 GHz - 22,2 GHz	Allocations: FIXED, MOBILE EXCEPT AERONAUTICAL MOBILE, RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE) Applications: Fixed, Radio astronomy, SRR, Video PMSE
22,2 GHz - 22,21 GHz	Allocations: FIXED Applications: Fixed, Radio astronomy, SRR, Video PMSE
22,21 GHz - 22,5 GHz	Allocations: FIXED, Mobile, MOBILE EXCEPT AERONAUTICAL MOBILE, RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE), Earth exploration-satellite (passive) Applications: Fixed, Radio astronomy, Video PMSE, SRR
22,5 GHz - 22,55 GHz	Allocations: FIXED, MOBILE, RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE) Applications: Fixed, Radio astronomy, Video PMSE, SRR
22,55 GHz - 23,15 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE, RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE) Applications: Fixed, Radio astronomy, Video PMSE, SRR

23,15 GHz - 23,55 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE Applications: Fixed, SRR, Video PMSE
23,55 GHz - 23,6 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE Applications: Fixed, Video PMSE, SRR
23,6 GHz - 24 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE) Applications: Passive sensors (satellite), Radio astronomy, SRR, Video PMSE
24 GHz - 24,05 GHz	Allocations: AMATEUR, AMATEUR-SATELLITE Applications: Amateur, Amateur-satellite, ISM, Non-specific SRDs, Video PMSE, SRR
24,05 GHz - 24,25 GHz	Allocations: Amateur, Earth exploration-satellite (active), Fixed, RADIOLOCATION, Mobile Applications: Active sensors (satellite), Amateur, ISM, LPR, Non-specific SRDs, Radiodetermination applications, SRR, TLPR, TTT, Video PMSE, Radiolocation (military)
24,25 GHz - 24,45 GHz	Allocations: FIXED, MOBILE Applications: Fixed, LPR, MFCN, TLPR, Video PMSE, SRR
24,45 GHz - 24,5 GHz	Allocations: FIXED, MOBILE Applications: LPR, MFCN, SRR, TLPR, Video PMSE, Fixed
24,5 GHz - 24,65 GHz	Allocations: FIXED, MOBILE Applications: Fixed, FWA, LPR, MFCN, TLPR, SRR
24,65 GHz - 24,75 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), MOBILE, FIXED Applications: Fixed, FWA, LPR, SRR, TLPR, MFCN
24,75 GHz - 25,25 GHz	Allocations: FIXED, FIXED-SATELLITE (EARTH-TO-SPACE), MOBILE, INTER-SATELLITE Applications: Fixed, FWA, LPR, MFCN, TLPR, SRR
25,25 GHz - 25,5 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE Applications: Aeronautical military systems, Fixed, FWA, Land military systems, LPR, MFCN, SRR, TLPR, Maritime military systems
25,5 GHz - 26,5 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE, SPACE RESEARCH (SPACE-TO-EARTH), Earth exploration-satellite (space-to-Earth) Applications: Aeronautical military systems, Fixed, FWA, Land military systems, LPR, Maritime military systems, MFCN, SRR, TLPR, Space research
26,5 GHz - 27 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE, SPACE RESEARCH (SPACE-TO-EARTH), Earth exploration-satellite (space-to-Earth) Applications: MFCN, Space research, SRR, TLPR, Land military systems
27 GHz - 27,5 GHz	Allocations: FIXED, INTER-SATELLITE, MOBILE, Earth exploration-satellite (space-to-Earth) Applications: Land military systems, MFCN
27,5 GHz - 28,5 GHz	Allocations: FIXED, FIXED-SATELLITE (EARTH-TO-SPACE), INTER-SATELLITE Applications: Feeder links, Fixed, FWA, GSO ESOMPs, NGSO ESOMPs, NGSO FSS, FSS Earth stations
28,5 GHz - 29,1 GHz	Allocations: FIXED, FIXED-SATELLITE (EARTH-TO-SPACE), INTER-SATELLITE, Earth exploration-satellite (Earth-to-space) Applications: Feeder links, Fixed, FSS Earth stations, GSO ESOMPs, NGSO ESOMPs, NGSO FSS, FWA
29,1 GHz - 29,5 GHz	Allocations: Earth exploration-satellite (Earth-to-space), FIXED, FIXED-SATELLITE (EARTH-TO-SPACE), INTER-SATELLITE Applications: Feeder links, Fixed, FWA, GSO ESOMPs, FSS Earth stations
29,5 GHz - 29,9 GHz	Allocations: Earth exploration-satellite (Earth-to-space), FIXED-SATELLITE (EARTH-TO-SPACE), INTER-SATELLITE, Mobile-Satellite (Earth-to-space) Applications: GSO ESOMPs, HEST, MSS Earth stations, NGSO ESOMPs, FSS Earth stations
29,9 GHz - 30 GHz	Allocations: EARTH EXPLORATION-SATELLITE (EARTH-TO-SPACE), FIXED-SATELLITE (EARTH-TO-SPACE), MOBILE-SATELLITE (EARTH-TO-SPACE), INTER-SATELLITE Applications: FSS Earth stations, GSO ESOMPs, MSS Earth stations, NGSO ESOMPs, HEST
30 GHz - 31 GHz	Allocations: FIXED-SATELLITE (EARTH-TO-SPACE), MOBILE-SATELLITE (EARTH-TO-SPACE) Applications: FSS Earth stations, MSS Earth stations, Satellite systems (military)
31 GHz - 31,3 GHz	Allocations: FIXED, MOBILE Applications: Fixed, Radio astronomy

31,3 GHz - 31,5 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE) Applications: Passive sensors (satellite), Radio astronomy
31,5 GHz - 31,8 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), Fixed, RADIO ASTRONOMY, SPACE RESEARCH (PASSIVE), Mobile except aeronautical mobile Applications: Fixed, Passive sensors (satellite), Radio astronomy
31,8 GHz - 32 GHz	Allocations: FIXED, RADIONAVIGATION, SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH) Applications: Fixed, FWA
32 GHz - 32,3 GHz	Allocations: FIXED, RADIONAVIGATION, SPACE RESEARCH (DEEP SPACE) (SPACE-TO-EARTH) Applications: FWA, Fixed
32,3 GHz - 33 GHz	Allocations: FIXED, INTER-SATELLITE, RADIONAVIGATION Applications: Fixed, FWA
33 GHz - 33,4 GHz	Allocations: FIXED, INTER-SATELLITE, RADIONAVIGATION Applications: FWA, Radiolocation (military), Fixed
33,4 GHz - 34,2 GHz	Allocations: RADIOLOCATION Applications: Radiolocation (military)
34,2 GHz - 34,7 GHz	Allocations: RADIOLOCATION, SPACE RESEARCH (DEEP SPACE) (EARTH-TO-SPACE) Applications: Radiolocation (military)
34,7 GHz - 35,2 GHz	Allocations: RADIOLOCATION, Space research Applications: Radiolocation (military)
35,2 GHz - 35,5 GHz	Allocations: METEOROLOGICAL AIDS, RADIOLOCATION Applications: Radiolocation (military), Active sensors (satellite)
35,5 GHz - 36 GHz	Allocations: EARTH EXPLORATION-SATELLITE (ACTIVE), METEOROLOGICAL AIDS, RADIOLOCATION, SPACE RESEARCH (ACTIVE) Applications: Active sensors (satellite), Radiolocation (military)
36 GHz - 37 GHz	Allocations: EARTH EXPLORATION-SATELLITE (PASSIVE), FIXED, MOBILE, SPACE RESEARCH (PASSIVE), Radio astronomy Applications: Passive sensors (satellite), Radio astronomy
37 GHz - 37,5 GHz	Allocations: FIXED, SPACE RESEARCH (SPACE-TO-EARTH) Applications: Fixed
37,5 GHz - 38 GHz	Allocations: FIXED, FIXED-SATELLITE (SPACE-TO-EARTH), SPACE RESEARCH (SPACE-TO-EARTH), Earth exploration-satellite (space-to-Earth) Applications: FSS Earth stations, Fixed
38 GHz - 39,5 GHz	Allocations: FIXED, FIXED-SATELLITE (SPACE-TO-EARTH), Earth exploration-satellite (space-to-Earth) Applications: FSS Earth stations, Fixed
39,5 GHz - 40 GHz	Allocations: FIXED, FIXED-SATELLITE (SPACE-TO-EARTH), MOBILE, MOBILE-SATELLITE (SPACE-TO-EARTH), Earth exploration-satellite (space-to-Earth) Applications: FSS Earth stations, MSS Earth stations
40 GHz - 40,5 GHz	Allocations: FIXED, FIXED-SATELLITE (SPACE-TO-EARTH), MOBILE, MOBILE-SATELLITE (SPACE-TO-EARTH), SPACE RESEARCH (EARTH-TO-SPACE), Earth exploration-satellite (space-to-Earth) Applications: MSS Earth stations, FSS Earth stations

7.6 Some sharing and compatibility studies already available

The studies listed below cover parts of the frequency bands 12 - 40 GHz for a range of low-power applications. While most date from many years previously, they may provide some useful information.

In ECC Report 139 [i.1] the impact of Level Probing Radars (LPR) using Ultra-wideband technology in all currently available frequency bands for LPR on radio communication services has been investigated. The studies in the frequency ranges around 24 GHz with the Fixed Service, EESS passive and radio astronomy could readily be used as reference for the higher frequency ranges requested in the present document.

ECC Report 023 [i.11] considers the impact of automotive Short Range Radars (SRR) on allocated radiocommunication services operating in the frequency range 21 - 27 GHz. The study focuses on the following three specific primary services to which SRR 24 GHz is considered likely to present a high interference potential:

- Fixed Service (FS)
- Earth Exploration Satellite Service (EESS)
- Radio Astronomy Service (RAS)

ECC Report 166 [i.12] elaborates relevant conditions under which meteorological radars, using zenithal pointing, should operate while ensuring coexistence with existing services, in view to provide a relevant background for future development and/or standardization of these radars.

An assessment of coexistence of security scanners with passive bands is available in ETSI EN 303 940-1 [i.13] (70 GHz scanners).

ITU-R reports covering passive bands include Report ITU-R SM.2092-0 [i.15] and Report SM.2091-0 [i.16].

7.7 Future studies

Coexistence is most likely to need verification with the following services:

- Earth exploration satellites, especially passive.
- Radio astronomy.

ECC Report 139 [i.1] considered interference issues 24 - 26,5 GHz and developed Table 6.

Table 6: Calculated LPR interference ranges to victim services in and near 24,05 - 26,5 GHz band

Victim	PP FS (civil)	PP FS (military)	PMP BS Rx	PMP TS Rx	RAS
Operating frequency, MHz	25 000	25 000	25 000	25 000	25 000
Bandwidth (IF), MHz	28	28	28	28	1
Receiver noise figure, dB	6	4,5	6	6	N/A
I/N objective, dB	-20	-20	-20	-6	N/A
Ga (in the direction of Interferer), dBi	41	47,4	18	0	0
Ga (sidelobes), dBi	0	0			
Feeder (insertion) loss, dB	0	1	3	0	0
Receiver thermal noise, dBm	**-93,36**	**-94,86**	**-93,36**	**-93,36**	**N/A**
Interference threshold at receiver input, dBm	**-113,36**	**-114,86**	**-113,36**	**-99,36**	**N/A**
Interference threshold before antenna, dBm	**-154,36**	**-161,26**	**-128,36**	**-99,36**	**-192,10**
Interference threshold PP FS side-lobes, dBm	**-113,36**	**-113,86**	**N/A**	**N/A**	**N/A**
LPR Interferer:					
Main beam mean e.i.r.p. limit (see ETSI TR 102 601 [i.18]), dBm (in the reference bandwidth)	20	20	20	20	-10*
Mean e.i.r.p. density, dBm/MHz	*-13,89*	*-13,89*	*-13,89*	*-13,89*	*-43,89*
Peak e.i.r.p. density, dBm/50 MHz	*26,10*	*26,10*	*26,10*	*26,10*	*6,10*
Antenna main-lobe gain, dBi	20	20	20	20	20
DeltaGa (first side-lobe, offset angle 20 - 30°), dB	-22	-22	-22	-22	-22
Antenna side-lobe gain > 60°, dBi	-10	-10	-10	-10	-10
Reflection loss, dB	13	13	13	13	13
Pulse Repetition Frequency, MHz	1,8	1,8	1,8	1,8	1,8
Reference bandwidth, MHz	2 450	2 450	2 450	2 450	2 450
Reflected e.i.r.p. in the direction of Victim, dBm	**-15**	**-15**	**-15**	**-15**	**-45**
Side-lobe e.i.r.p. in the direction of Victim, dBm	**-10**	**-10**	**-10**	**-10**	**-40**
Total interfering power towards Victim, dBm	**-8,81**	**-8,81**	**-8,81**	**-8,81**	**-38,81**
Horizontal power density check, dBm/MHz (e.i.r.p.)	*-42,70*	*-42,70*	*-42,70*	*-42,70*	*-72,70*
Conditional BW Correction Factor, dB	**-19,42**	**-19,42**	**-19,42**	**-19,42**	**-31,34**
Impact range calculation:					
Minimum Coupling Loss balance, dB	**126,13**	**133,03**	**100,13**	**71,13**	**121,95**
Impact range with FSL model, km	**1,935**	**4,282**	**0,097**	**0,003**	**1,196**
Impact range calculation (in PP FS side-lobes):					
Minimum Coupling Loss balance, dB	**85,13**	**85,63**	**N/A**	**N/A**	**N/A**
Impact range with FSL model, km	**0,017**	**0,018**	**N/A**	**N/A**	**N/A**

This showed that with a 20 dBm EIRP main beam (except for radio astronomy), interference ranges were up to 4 km. However, this was for a mixed indoor/outdoor deployment (90 % indoor, 10 % outdoor). In such scenarios outdoor interference tends to be the limiting factor. With 100 % security scanners indoors the additional building penetration losses, interference ranges are likely to be well below the dimensions of the building in which the scanner is located.

For the passive EESS bands the study showed a maximum transmit power between -18 dBm/MHz and -21 dBm/MHz depending on the victim service. Again this was for a mixed indoor/outdoor interferer.

7.8 Information on relevant standards

Table 7: Information on relevant standards

Application	Frequency Ranges [GHz]	ETSI Standard	Status	Remark	Responsible ETSI TC ERM
Short Range Devices (SRD)	40 - 246 GHz	ETSI EN 305 550-1 [i.5] ETSI EN 305 550-2 [i.7]	EN Approval Procedure (ENAP) started	RED compliant	TGUWB
Tank Level Probing radar (TLPR)	4,5 - 7 GHz, 8,5 - 10,6 GHz, 24,05 - 27 GHz, 57 - 64 GHz, 75 - 85 GHz	ETSI EN 302 372 [i.2]	Cited in the OJEU	RED compliant	TGUWB
Level Probing Radar (LPR)	6 - 8,5 GHz, 24,05 - 26,5 GHz, 57 - 64 GHz, 75 - 85 GHz	ETSI EN 302 729 [i.3]	Cited in the OJEU	RED compliant	TGUWB
Security Scanner (SSc)	69,8 GHz - 80,5 GHz	ETSI EN 303 940-1 [i.13]	EN		TGUWB

8 Radio spectrum request and justification

The use case considered in the scope of the present document will be qualified as fixed or quasi-fixed radiodetermination applications as SScs are not very portable or are required at a certain location of interest. An SSc will typically need to be taken down before it can be moved to a different location. In an installed setting, it cannot be moved.

The security checkpoint application is deployed indoors in thermally efficient buildings. The proposed limits, bandwidth and frequency range are given in Table 8 below.

Table 8: Proposed limits for maximum peak power and frequency range

Frequency Band [GHz]	Power/Magnetic Field
12 - 15,35	15 dBm Peak EIRP in the direction of main radiation
15, 4 - 23,6	15 dBm Peak EIRP in the direction of main radiation
24 - 31,3	15 dBm Peak EIRP in the direction of main radiation
31,5 - 40	15 dBm Peak EIRP in the direction of main radiation
NOTE: No intentional (wanted) emissions in 15,35 - 15,4 GHz, 23,6 - 24 GHz, 31,3 - 31,5 GHz according to received LS in ECC(22)039 [i.17], Annex 16.	

These power levels compare to those in other SSc bands as follows:

Table 9: Comparison with other SSc power limits

Band	Peak EIRP
3 - 10 GHz	0 dBm / 50 MHz
12 - 40 GHz	15 dBm
70 - 80 GHz	7 dBm (19 dBm / 76 - 80 GHz)

Higher powers are needed in the 12 - 40 GHz band because the intent is to enable a faster flow through the scanner, allowing for constant motion (e.g. walking through, rather than stopping during the scan), allowing for thicker clothing to be worn rather than having to be taken off and scanned separately, and also scanning through footwear, again removing the need for this to be taken off and scanned separately. This faster flow is very valuable to those operating the scanners and results in greater convenience and shorter queues for those being scanned, enabling, for example, shorter queuing times at airport security. These are very important outcomes that merit the request for higher power levels.

Note that not all scanners will use the entire range 12 - 40 GHz (excluding the passive bands - see Table 8). For example, some might scan 20 - 40 GHz.

9 Regulations

9.1 Current regulations

There are currently no SSc frequency designations from 12 GHz to 40 GHz.

9.2 Proposed regulation

Table 10: Proposed regulatory parameters (in dependence on ERC/REC 70-03 [i.9], Annex 6)

Frequency Band [GHz]	Power/Magnetic Field	Spectrum access and mitigation requirements	Modulation/ maximum occupied bandwidth	Usage
12 - 15,35	15 dBm Peak EIRP in the direction of main radiation	Not specified	Not specified	For indoor usage of SSc
15, 4 - 23,6	15 dBm Peak EIRP in the direction of main radiation	Not specified	Not specified	For indoor usage of SSc
24 - 31,3	15 dBm Peak EIRP in the direction of main radiation	Not specified	Not specified	For indoor usage of SSc
31,5 - 40	15 dBm Peak EIRP in the direction of main radiation	Not specified	Not specified	For indoor usage of SSc
NOTE: No intentional (wanted) emissions in 15,35 - 15,4 GHz, 23,6 - 24 GHz, 31,3 - 31,5 GHz according to received LS in ECC(22)039 [i.17], Annex 16.				

History

Version	Date	Status
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